

SCREEN TIME AND MENTAL HEALTH PROBLEMS

A population-based study [SALVe] about screen time contribution to mental health problems among adolescents in Västmanland

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ABSTRACT

There is an increasing trend of mental health problems both globally and in Sweden. Moreover, in recent decades there has been an increase in screen time among adolescents. The present study aimed to examine the associations between screen time (i.e., smartphone, computer, and TV) and mental health problems among adolescents in Västmanland and to investigate if the association was different due to gender. A quantitative method with a cross-sectional design was applied. The study used secondary data from the Survey of Adolescent Life in Västmanland 2020. The sample consisted of 3880 adolescents from 9th grade in compulsory school and 2nd grade in upper secondary school.

The results showed that high screen time on smartphone was associated with an increased probability for mental health problems in the total sample. In contrast, screen time on TV and computer showed no significant association with mental health problems. Thus, smartphone use was a significant contributor to mental health problems. Furthermore, for girls, high screen time on the smartphone, computer, and TV was associated with increased probability of mental health problems, whereas no significant associations were found among boys. In brief, this study's findings suggest developing Swedish guidelines to regulate harmful effects from screen time.

Keywords: *Anxiety, Depression, Screen types, Smartphone, The displacement hypothesis, Public health*

CONTENTS

1	INTRODUCTION	1
2	BACKGROUND.....	2
2.1	Mental health problems among adolescents.....	2
2.2	Recommendations for Screen time.....	3
2.3	The association between screen time and mental health among adolescents...	4
2.3.1	<i>Gender differences in the association</i>	<i>5</i>
2.4	Theoretical framework- The displacement hypothesis.....	6
2.5	Problem formulation.....	7
3	AIM AND RESEARCH QUESTIONS.....	7
4	METHOD AND MATERIAL	7
4.1	Methodological approach and Study design	8
4.2	Sample and data collection.....	8
4.3	Measurements	8
4.3.1	<i>Dependent variable.....</i>	<i>9</i>
4.3.2	<i>Independent variables.....</i>	<i>9</i>
4.3.3	<i>Confounders</i>	<i>10</i>
4.4	Statistical Analysis	10
4.5	Research Ethics.....	11
5	RESULTS	12
5.1	Associations between screen time and mental health problems.....	12
5.2	Associations between screen time and mental health problems in girls and boys.....	13
6	DISCUSSION.....	15
6.1	Method discussion	15
6.1.1	<i>Sample and data collection</i>	<i>15</i>
6.1.2	<i>Measurements and statistical analysis</i>	<i>16</i>
6.1.3	<i>Ethical considerations</i>	<i>18</i>

6.2 Results discussion.....	18
6.3 Relevance to public health and future research.....	21
7 CONCLUSIONS	22
REFERENCE LIST	23
APPENDIX A; QUESTIONNAIRE, SALVE 2020	

1 INTRODUCTION

Mental health problems such as depression and anxiety are current public health issues. For instance, World Health Organization (2017) claim that about 322 million individuals have been diagnosed with depression globally, and corresponding number for anxiety are 264 million people worldwide. Furthermore, on average, females suffer more than males from mental health problems (WHO, 2017). Moreover, there is an ongoing trend showing an increase in mental health problems among adolescents (Choi, 2018). Globally, it has been estimated that approximately 10-20% of the adolescent population suffer from mental health problems (WHO, 2021a). Mental health problems are commonly developed during adolescence, and there are several risk factors for mental problems (Erskine et al., 2015). For example, is low social support a risk factor, and substance use is a risk behaviour that can negatively affect mental health. In addition, extensive social media use is associated with decreased mental well-being (Walsh et al., 2020). However, it is not clear why mental health problems such as depression among youth are increasing (Folkhälsomyndigheten, 2018). Although there are many proposed explanations for the growing trend, one of them may be the digital development.

Many activities include using screens in society today, e.g., people use smartphones frequently, using computers and looking at the TV. This leads to increased screen time in the population. Many Swedish children have grown up in a digital society, and youth are more exposed to different screens than the previous generations. Indeed, the various screens offer entertainment such as games and digital interactions (Liu et al., 2016). Thomée (2018) claims that computers and tablets are used daily, both at school and in leisure time. However, it is the arrival of the smartphone that has changed the way we live. Smartphone use has changed how we communicate and interact with each other. Therefore, it has resulted that adolescence is more devoted to screen-activities, which can cause other important aspects in life to be overlooked, such as social interactions with peers, studying and physical activity (Thomée, 2018). In the past, the TV was the dominant contributor to screen time. However, nowadays, it has changed, and smartphones are the most frequent contributor to screen time, especially among adolescents (Saunders & Vallance, 2017).

The relationship between screen time and mental health problems has been noticed. Although, few studies have assessed how each type of screen influences mental health (Stiglic & Viner, 2019). In sum, there is an increase in mental health problems among adolescents. At the same time, an increasing trend in screen time has also been noticed. Therefore, a potential explanation for adolescents' mental health problems may be related to screen time. Further research is needed to examine associations of different types of screens and to understand which screen type is most harmful. From a public health perspective, it is interesting to deepen the knowledge about how screen time impact mental health among youth. This knowledge can be helpful for the development of interventions to improve mental health and reduce screen time.

2 BACKGROUND

Mental health problems constitute a heavy burden for society. However, there is still a lack of evidence for underlying factors that can lead to mental health problems (Erskine et al., 2015). Mental health is also of interest concerning the sustainable development goals in Agenda 2030. Sustainable development goal 3 in Agenda 2030 covers good health and well-being and intending to promote mental health through preventative interventions (WHO, 2021b). In the present study, the interest was to study how screen time contributes to mental health and how adverse mental health outcomes could be prevented by increasing the understanding of potential risk factors for mental health problems.

2.1 Mental health problems among adolescents

Mental health as concept includes several aspects to consider. According to Choi (2018), is mental health affected by several factors, and together they constitute a complex interplay. This contributes to the fact that the concept of mental health is multidimensional (Choi, 2018). Bremberg and Dalman (2015) argue that there is a debate about what mental health exactly consists of, such as the absence of illness is not equal to be mentally healthy. For instance, mental health includes several dimensions, such as psychological, social, and emotional dimensions. Furthermore, mental health problems can be explained as a comprehensive concept, including minor and severe symptoms, such as depressive and anxiety symptoms, without focusing on clinical diagnosis (Bremberg & Dalman, 2015).

In the present study, mental health problems were defined as anxiety- and depressive symptoms. The concept will be measured with the psychological distress scale developed by Kessler et al. (2002). It is a measurement for non-specific psychologic distress, which can be treated as potential diagnosable anxiety and depression (Kessler et al., 2002). Depression can for example be expressed as sadness, tiredness, and powerlessness, whereas anxiety includes different feelings of fears and nerviness (WHO, 2017).

Beyond risk factors such as screen time, other factors may also contribute to mental health problems. For instance, previous researchers (e.g., Hamer & Stamatakis, 2014) has identified sedentary as a risk factor for mental illness. Sedentary behaviours have been found to impact mental health negatively. According to Kandola et al. (2020) is sedentary associated with depressive symptoms. Besides, sedentary behaviour often increases during adolescence (Kandola et al., 2020). In addition, screen activities such as internet use and watching TV is related to sedentary behaviours, and such activities are also common among adolescents (Hamer & Stamatakis, 2014). Moreover, another well-established risk factor for mental health problems is low socioeconomic status (McLaughlin et al., 2012; Reiss, 2013). In addition, it exists associations between low socioeconomic status and extensive screen time (Allen & Vella, 2015). Therefore, the present study adjusted for these potential risk factors.

Adolescence is a sensitive period where many changes in the body take place, both physically and mentally. It includes the development of several important cognitive and emotional functions. It is common that mental health problems occur for the first time at the age of 14

(Ogden & Amlund Hagen, 2019). According to Bor et al. (2014), there has been an increase in mental health problems among adolescents globally, where girls report more mental health problems (Bor et al., 2014). In the Swedish context same trends has identified. Swedish adolescents also reporting increase in mental health problems, yet the increasing trend is to some extent unexplained (Bremberg, 2015). It is more common that girls suffer from anxiety, nervousness, or anxiousness than boys. For example, in a Swedish study, 29 % of the girls reported impaired mental well-being, while 7 % of the boys reported impaired mental well-being. Notably, adolescents aged 16-19 years reported most mental health problems (Lager et al., 2012).

2.2 Recommendations for Screen time

Today, digitalization is a natural part of children and adolescents' lives. Due to technological development, televisions, smartphones, computers, and tablets are screen types that are daily used. Digital usage can be described as screen time, which is the time spent in front of the screens (Rosen et al., 2014). Digital media usage has become a big part of our lives and contributes to increasing time spent in front of screens (Stiglic & Viner, 2019). During the 21st century, there has been an extensive rise in media usage among Swedish adolescents (Folkhälsomyndigheten, 2018), mainly related to smartphones' use (Thomée, 2018). In addition, screens like smartphones and tablets are portable, which make them always accessible compared to the TV, which is stationary (Twenge et al., 2019).

It exists some general recommendations regarding the amount of acceptable screen time among adolescents. Tremblay et al. (2016) suggest a maximum of two hours per day in leisure time for adolescents. Saunders and Vallance (2017) also argue for a maximum of two hours of daily screen time among children and youth between five to 17 years old. It is important to reduce the amount of time youth spend in front of screens to prevent adverse health outcomes (Saunders & Vallance, 2017). In the long term, extensive amount of screen time may be harmful to mental health.

When the recommendations of daily screen time are exceeded, it may impact mental health negatively. Twenge and Campbell (2018) study showed associations between screen time and reduced mental health among adolescents. In particular, decreased mental well-being was noted among adolescents with at least one hour of daily screen time. Moreover, the probability to have been diagnosed with anxiety or depression was more than doubled among adolescents (14 to 17 years old) with screen time of seven or more hours per day compared to those with one hour of screen time. Screen time was measured on an average weekday of time spent watching TV, computer- and mobile phone use. In summary, the study revealed the total screen time of over one hour per day increased the risk of mental health problems (Twenge & Campbell, 2018). Furthermore, when screen time was summarized as computer games and tv-watching in leisure time on a weekly average, the results showed that adolescents with screen time over two hours per day had an increased risk of mental health problems (Herman et al., 2015).

2.3 The association between screen time and mental health among adolescents

Screen time is related to adolescent's mental health. Stiglic and Viner (2019) meta-analysis showed evidence for associations between screen time and depressive symptoms. However, most of the included studies only measured screen time in front of the television. Thus, it is not possible to conclude that screen time, in general, contributes to mental health problems. More research is needed to broaden the understanding of the association, e.g., by including several screen devices. Therefore, more evidence is needed to draw valid conclusions (Stiglic & Viner, 2019). Cross-sectional results found that the combined screen time on computer, TV and video gaming was associated with various mental health problems. Different types of screen activities, i.e., computer and video games, were related to mental health problems such as anxiety- and depressive symptoms (Maras et al., 2015).

Liu and colleagues (2016) meta-analysis reveal that adolescent with high screen time from computer use, and TV watching had an increased risk for depression. Among youth with over two hours of daily screen time increased the risk for depression compared to the reference group with no or less than two-hour screen time. The included studies did not evaluate the effects of all devices on screens. Hence, how smartphones and tablets contribute to mental health was not measured (Liu et al., 2016). In sum, this indicates that watching TV can influence mental health negatively (Stiglic & Viner, 2019; Liu et al., 2016). Nevertheless, how other types of screens impact mental health could be further investigated.

Today is the smartphone the most common screen type among youth. Thomée (2018) argues that a high amount of screen time on the smartphone is associated with a negative contribution to mental health. Among both children and adolescents were extensive smartphone use related to depressive symptoms (Thomée, 2018). Twenge et al. (2018) state that adolescents with the lowest screen time have better mental health status. This study measured TV watching and electronic communication, for instance, texting and social media, separately. The results showed that watching TV was related to decreased mental well-being. Electronic communication such as gaming and social media was associated with decreased happiness and reduced satisfaction with life (Twenge et al., 2018). Hence, the cross-sectorial results indicate that adolescents spending a lot of time in front of screens tend to report reduced mental health. However, since screen time is a broad concept, the studies have measured it in several ways.

Over time, screen time as a concept has been more complex. It also seems to be a change in the use of screens since there are many devices. Domingues-Montanari (2017) argues that TV is the most common screen type among young children, which changes in adolescence where smartphones, tablets, and computers account for the most common screen types. Because of many technology devices, they may contribute and affect in different ways (Domingues-Montanari, 2017). Twenge and Campbell (2018) claim that it is important to measure screen time, including several screen types. In addition, few studies have studied the relation between screen time and psychological well-being (Twenge & Campbell, 2018). Screen time may be less harmful depending on the screen type. Rosen et al. (2014) argue that depending on the type of screen, it can impact mental health in different ways. Hence, watching TV may

be less harmful than engaging in computer games and social media usage. The risk of mental problems is most significant when face-to-face interaction is replaced by digital interactions (Rosen et al., 2014). Further research is required to examine how different types of screens contribute to mental health. In the present study, screen time will be defined as smartphone use, TV-watching, and computer (tablet) use during leisure time.

Although previous research primarily found a negative impact from screen time on mental health, there are also positive aspects with the development of the technology. Some screen types offer social interaction, which some authors state to be positive for mental health. The shift regarding the usage of screens has made smartphones the most attractive type of screen to use among adolescents. Some authors affirm that screen time contributes to mental well-being. For instance, Przybylski and Weinstein (2017) study indicate positive effects of screen time on mental health. Screen time was measured as smartphone use, computer use, and watching TV. Screen time itself was not harmful to adolescent's mental health. This study indicates that average screen time does not necessarily have to be harmful to mental health. However, it is still possible that high amounts of screen time may influence mental health in a negative way (Przybylski & Weinstein, 2017). Another study argues that playing video games can improve mental health and contribute to psychosocial benefits since online gaming offers social interaction. It creates social meetings and possibilities to meet new friends who can positively affect mental health. However, screen time was assessed only through video gaming (Granic et al., 2014). Additionally, there appears to be a positive relationship between online communication and mental well-being. Nowadays, there are digital forums that, for example, offer social support through online contacts. This can broaden social capital and self-confidence, contributing to positive health effects (Best et al., 2014).

In conclusion, previous research has revealed both positive and negative associations between screen time and mental health. The contradictory results may be due to the type of screen the adolescents use, as each screen type and time spent using screens has been measured differently. Accordingly, Twenge and Farley (2021) claim that the results may differ due to the different ways screen time has been assessed. Thus, there is a lack of evidence regarding each screen type impact on mental health.

2.3.1 Gender differences in the association

It is also of interest to determine if screen time contributes to mental health in the same way among boys and girls. Oberle et al. (2020) study showed that screen time from watching TV and computer games of at least two hours in leisure time was associated with mental health problems among both girls and boys. However, the association was stronger among girls. Hence, more research is motivated to explore why screen time may impact girl's mental health to a wider extent than boys. Furthermore, Twenge and Farley (2021) state that the association can differ depending on the type of screen and gender. Cross-sectional results found girls with extensive screen time on the internet and social media usage were the strongest predictor for reduced mental health. The association was much stronger for girls than for boys and the association was different depending on the screen type. In conclusion, this means that not all types of screens can be considered equal. The strongest association for

mental health problems was related to social media usage, while watching TV and computer games were weaker. The association was found in both genders, even if it was stronger for girls. Consequently, future research where gender is used as a moderator is needed (Twenge & Farley, 2021). With that in mind, more attention is motivated to examine gender differences since screen time seems to affect girl's mental health negatively more than boys. Especially, smartphone usage indicates to contribute to mental health problems, particularly among girls.

2.4 Theoretical framework- The displacement hypothesis

A commonly used theory to explain the relationship between screen time and adolescents' mental health is the displacement hypothesis. The theory suggests that the time spent in front of screens displaces time, for instance, health promotion activities. Thus, screen time can negatively affect adolescent's mental health. The theory is derived from Neuman (1988) that state technology such as watching TV influences mental health negatively. Based on this theory, the present study hypothesized that screen time might decrease mental health. The time adolescents spend on screens reduces the time that instead could be spent on, for example, physical activity or social meetings.

Furthermore, it is the replacement of activities with positive effects for an individual's life and health that contribute to adverse health outcomes. In other words, the negative impact on mental health is due to neglecting other activities, for example, sports, doing homework, and social interactions with family and friends. Therefore, screen time can be harmful to mental health. The negative impact occurs significantly when important health-promoting activities are displaced (Neuman, 1988). Kraut et al. (1998) also confirm the displacement hypothesis. When screen time is too extensive, the author states there is a risk for interpersonal relationships to be suffered (Kraut et al., 1998).

A limitation of the theory is that it was formed when screen time primarily constituted of watching TV. Over time the activities related to screen time have changed, and today the smartphone, for example, offers a variety of social interactions. Hence, the type of activity associated with each particular screen type is nowadays different. For instance, the smartphone might contribute to social interactions while the TV contains of passive acts. Furthermore, in previous studies screen time have been assessed in several ways, which may have caused contradictory results. Depending on the conceptualization, screen time may contribute in different ways. Some studies measure the totally screen time from several screens, and others only include a specific screen type contribution. Liu et al. (2016) argue that the screen type is a significant moderator for the association (Liu et al., 2016).

Based on previous knowledge screen time constitute to be a possible risk factor for mental health problems among adolescents. Consequently, it is of interest to further investigate how screen time contribute to adolescent's mental health. In the present study, the interest is directed towards examining respective screen type contribution. Thus, the different screen types are examined separately. Screen time in the present study will consequently be defined as TV, smartphone, and computer (including tablet). Based on existing knowledge about the

association between screen time, mental health, and the displacement hypothesis, it is hypothesized that the three different screen types influence mental health negatively.

2.5 Problem formulation

In the field of public health, it is interesting to identify risk factors to prevent the onset of mental health problems. Previous research indicates an increased risk for mental health problems among adolescents with extensive screen time. However, there is a lack of studies investigating which particular screen type constitutes an increase in mental health problems. Therefore, it is a priority to determine the impact of different screens to prevent a negative development of mental health. In this study, the interest is to investigate how screen time affects mental health among a sample of adolescents in Sweden. Specifically, the present study aims to examine the contribution of three different screen types (smartphone, computer, and TV) to mental health problems. In addition, determine if some particular screen type is more harmful to mental health. There is a lack of adequate research on gender differences for the association between screen time and mental health problems. Therefore, the present study will examine the association separately for girls and boys.

3 AIM AND RESEARCH QUESTIONS

The aim was to examine the associations between screen time (i.e., smartphone, computer, and TV) and mental health problems among adolescents in Västmanland and to investigate if the association was different due to gender.

1. Is there an association between the different screen types and mental health problems among adolescents?
2. Which screen type is most associated with mental health problems?
3. Is the association different for girls and boys?

4 METHOD AND MATERIAL

The methodological approach, sample procedure, and statistical analysis of the present study are described in the followings section.

4.1 Methodological approach and Study design

The present study aims to examine associations between screen time and mental health problems. The quantitative method is considered suitable, since it allows for investigating relationships between several variables, which was in line with Creswell and Creswell (2018) that claim a deduction approach is when theories form the foundation for the research questions. Hence, the quantitative method is appropriate for answering the present study research questions and exploring the association.

The present study used secondary data from the Survey of Adolescent Life in Västmanland 2020 [SALVe]. The county council of Västmanland conducted the survey and is responsible for the data collection. It is a comprehensive population survey with a cross-sectional design (Söderqvist & Larm, 2021). Consequently, the current study design is cross-sectional, and the participants were only observed at one time.

4.2 Sample and data collection

The present study used secondary data from SALVe 2020. According to Söderqvist and Larm (2021) is the survey conducted by the county council of Västmanland and collects information about adolescents living in Västmanland life, health, and life habits. The data was collected in the early spring of 2020. The participants completed the questionnaire in a classroom environment during school time. The students filled out a web survey individually during examination similar conditions (Söderqvist & Larm, 2021).

The sample in the present study consisted of adolescents from 9th grade of compulsory school and second grade from upper secondary school in the county of Västmanland (Söderqvist & Larm, 2021). The SALVe survey has been conducted since 1995 with the purpose to follow the health development in the county and contribute with a decision basis for further interventions (Region Västmanland, 2020). All adolescents from the target population were invited to participate in the survey. The participants were from both municipal and private schools. The study population consisted of 5480 adolescents from the whole county (Söderqvist & Larm, 2021). In the present study, the final sample consisted of 3880 participants, including a response rate of 71 %. Of the sample 49.7 % were boys (n=1930) and 50.3% were girls (n=1950).

4.3 Measurements

The studied variables are described below. Selected items from the SALVe 2020 questionnaire are presented in Appendix A.

4.3.1 Dependent variable

In the present study, the outcome of interest was mental health problems, and the concept are defined as anxiety- and depressive symptoms. Mental health problems were measured with a self-reported questionnaire containing of sixth items measuring psychological distress. The psychological distress scale K6, developed by Kessler et al. (2002), was used to assess the probability for potential anxiety- and depressive symptoms. It is a measurement that estimates individuals at risk for psychological distress in the studied population (Kessler et al., 2002). The questionnaire (Appendix A) consists of questions about; How often during the last month have you felt *“nervous, hopeless, restless or fidgety, so depressed that nothing could cheer you up, everything is an effort, and worthless.”* The psychological distress scale consists of a five-point frequency scale with the response alternatives: from 0 to 4. *“None of the time”, “a little of the time”, “some of the time”, “most of the time”, or “all of the time”*. Hence, the scale has a range from 0-24 when the six items are summed up. Higher scores on the scale indicate higher levels of psychological distress which can be treated as anxiety and depression. In the present study, an index was created of the 6 items from the psychological distress scale. The internal non-response for the index was n= 341, 8.8 %. According to Kessler et al. (2002), is the Cronbach’s α for the original scale 0.89, which mean that both reliability and internal consistency are very good (Kessler et al., 2002). In the present study, Cronbach’s α value for the index of the sixth items was 0.91.

Several population-based health surveys have shown the screening instrument to be accurate to measure serious psychological distress (Veldhuizen et al., 2007). Kessler et al. (2003) claim that the original 6-item psychological distress scale with the cut-off ≥ 13 has a sensitivity of 0.36, specificity of 0.96, and classification accuracy of 0.92 (Kessler et al., 2003). According to Mitchell and Beals (2011), is the cut-off ≥ 13 widely used and acceptable to indicate an increased likelihood of serious mental illness. This means a higher probability of have a diagnosable mental illness (Mitchell & Beals, 2011). Further, the index was dichotomized to identify potential anxiety- and depressive symptoms. The established cut-off ≥ 13 was used to create a binary variable. Hence, the index was dichotomized into <13 “No mental health problems” and >13 “Mental health problems”.

4.3.2 Independent variables

In the present study, the independent variables are screen time, and it was measured with a self-reported questionnaire. In the SALVe 2020 survey, several questions were about screen time. For the current study, three questions were selected to represent three different types of screen time namely “On average, how many hours per day do you use a computer/tablet in your leisure time (not at school)?” The internal non-response for computer/tablet in leisure was n= 668, 16%. Further computer and tablet will be defined as only “computer” even if it includes tablets to. “On average, how many hours per day do you watch on TV/ TV programs (including streaming services)?” The internal non-response for TV was n=677, 17 %. “On average, how many hours per day do you use your smartphone?” The internal non-response for smartphone was n= 675, 16.9%. The response options ranged from 0 to more than 10 hours per day. “Mark with a ring on the timeline to show the number of hours.” In the present study, the variables for screen time were dichotomized according to previous studies

(Shiue, 2015; Duncan et al., 2012). Hence, high screen time was categorized as > 4 h per day and low screen time as < 4 h per day for each screen type.

In the present study, gender was used as a moderator. Field (2018) states that a moderator can impact the association between a predictor and an outcome. To understand if the association differs between girls and boys, the sample was stratified using gender as a potential moderator.

4.3.3 Confounders

It is important to control the association for potential confounders. Based on previous research on screen time and mental health problems, the potential confounders in the present study are sedentary status and perceived socioeconomic status (Hamer & Stamatakis, 2014; Reiss, 2013). The presented confounder variables may influence the association and will therefore be adjusted for in the analysis. Allen and Vella (2015) claim that adolescents having low educated parents or low socioeconomic status are more likely to have a high amount of screen time (Allen & Vella, 2015). In addition, low socioeconomic status is a risk factor for mental health problems. There is an increased risk of developing various mental health problems among adolescents with low SES (Reiss, 2013). In the present study, perceived socioeconomic status was assessed with the question: *"Imagine the society as a ladder. If you think about your family's economy compared to society at large, where would you place your family on the steps?"* The response options were between 1 to 7. 1) At the bottom, least money, 7) At the top, the most money (M= 4.58, SD= 1.07).

The other potential confounder in the present study is sedentary status. Sedentary status is associated with an increased risk for developing depressive symptoms and can negatively impact mental health (Hamer & Stamatakis, 2014). Sedentary is also closely linked to screen time, such as watching tv and computer use (Tremblay et al., 2011). Therefore, it was of interest to control the eventual impact from sedentary, to conclude if mental health is derived from screen time rather than sedentary behaviour. Sedentary status was measured with the question: *"How much do you sit during a normal day?"* The response options were in a range from 1 to 4. Less than 4 hours, 4-6 hours, 7-9 hours, and 10 hours or more (M= 2.71, SD= 0.81).

4.4 Statistical Analysis

The analyses were conducted using the IBM SPSS Statistics (version 26). Initially, descriptive statistics were performed to see the frequency distribution of the included variables. In the descriptive analyses, histograms were used to investigate the distribution of the included variables. The index of mental health problems was found not to be normally distributed. Hence, the logistic regression was considered appropriate since the analysis does not require normal distribution of the dependent variable. Furthermore, the dichotomized physiological distress variable was used in the analysis.

To perform logistic regression, it is of importance to check the multicollinearity. According to Field (2018), there is a risk for bias with strong correlations between independent variables. Consequently, control for multicollinearity is an assumption in logistic regression (Field, 2018). In the present study, multicollinearity was controlled with Pearson correlation. However, in the analysis, there was no risk for multicollinearity problems. The highest correlation was between computer and TV ($r=.374$, $p<.001$). According to Field (2018), a correlation of $>.70$ is a concern for multicollinearity. Therefore, the variables were suitable for the analysis.

The logistic regression began with multivariate unadjusted analyses, which help to examine how the odds ratios change of eventual impact from the potential confounders. The next step was to conduct the regression, including potential confounders. The multivariate logistic regression estimates each of the predictor's contribution to mental health problems adjusted for confounders. Hence, all three screen types were included simultaneously with confounders. In the end, the multivariate analyses were conducted with a stratified sample. First unadjusted and then adjusted to determine if the association were different for girls and boys. Consequently, to answer the last research question, analyses were conducted with the sample stratified into boys and girls.

4.5 Research Ethics

Before the adolescents answered the questionnaire, a video was shown. The video informed the participants that it was voluntary to participate and that they could quit whenever they want without giving a reason, and their participation was anonymous. They also received information about the aim of the study before they logged in to answer the survey. Consent to participate was collected through the students completing the survey. The researcher did not collect any information about the participants' social security number or names. Thus, all participants were preserved anonymously. The study's ethics meet the Declaration of Helsinki. In addition, ethical approval was given from the ethics review authority (Dnr 2019–05620) (Söderqvist & Larm, 2021). The questionnaires were inevitably kept from unauthorized persons. Only people who analyse the results has access to the data. Further, the adolescents have received written information that addressed the purpose and utility of the survey (Region Västmanland, 2021). Consequently, the responsible for the data collection is thought to have considered the information and consent requirements.

An agreement was signed with the county council of Västmanland regarding confidentiality in the handling of the data set. The data set only included the parts from SALVe 2020 needed to answer the aim of the present study. The data set was released with a cloud link with a password. The use of the data was limited to be used during the spring of 2021 and for the specific topic of the present study (screen time and mental health problems). The dataset and its contents belong to the county council of Västmanland, and confidentiality for the content applies both during and after the study. The data consisted of only needed variables to be able to answer the study's aim and only relevant information for the topic is allowed to disseminate. All other information from the dataset is covered by statistical confidentiality.

5 RESULTS

The included variables descriptive statistics are presented below. Initially, the frequency and percentages of the categorical variable for mental health problems are described (*Table 1*). In the total sample, the majority had no mental health problems (82.5%). However, 17.5 % had potential diagnosable mental health problems. It was more common among girls (24.2 %) than boys (10.3%) to have mental health problems.

Table 1: Dichotomized variable; No mental health problems and Mental health problems, presented in number and percent for girls (N=1950) and boys (N=1930) separately and the total sample (N=3880).

	No mental health problems	Mental health problems
Girls	n=1379, 75.8 %	n=441, 24.2 %
Boys	n=1542, 89.7%	n=177, 10.3%
Total sample	n=2921, 82.5 %	n=618, 17.5 %

It was shown that the highest screen time in the total sample was on smartphones (65.2 %) (*Table 2*). Among boys, the highest screen time was found from the computer (65.4%), while girls had the highest screen time on the smartphone (73 %).

Table 2: Dichotomized variables smartphone, computer, and TV. Number and percent for low and high screen time. Girls (N=1950) and boys (N=1930) separately and in the total sample (N=3880).

	Girls		Boys		Total Sample	
	Low	High	Low	High	Low	High
Smartphone	n=449, 27%	n=1215, 73%	n=670, 43%	n=885, 57%	n=1119, 34.8%	n=2100, 65.2 %
Computer	n=825, 49.3%	n=847, 50.7%	n=538, 34.6%	n=1016, 65.4%	n=1363, 42.3%	n=1863, 57.7%
TV	n=1044, 62.7%	n=621, 37.3%	n=907, 58.4%	n=646, 41.6%	n=1951, 60.6%	n=1267, 39.4%

Notes: Dichotomized variables; Low screen time <4 h per day and high screen time >4 h per day.

5.1 Associations between screen time and mental health problems

The unadjusted results for the total sample (*Table 3*) found the association between smartphone and mental health problems statically significant ($OR=1.61$, $CI\ 95\ % = 1.30-1.99$,

$p < .001$). Hence, high screen time on smartphone increases the odds by 61% for having mental health problems compared to those with low screen time. In addition, high screen time from TV ($OR=1.29$, $CI\ 95\ \% = 1.05-1.58$, $p=.014$) was significant. High screen time from watching TV increases the odds by 29 % for having mental health problems. However, no significant results were found for the computer ($OR=1.07$, $CI\ 95\ \% = 0.87-1.31$, $p=.51$) in the total sample.

When the regression was adjusted for the potential confounders (sedentary status and perceived socioeconomic status) (Table 3), the only statically significant screen type associated with mental health problems was the smartphone ($OR=1.72$, $CI95\%= 1.39 - 2.13$, $p < .001$). Hence, adolescents with high screen time on smartphone had increased odds by 72 % for diagnosable mental health problems compared to those with low screen time. However, no significant association was found among screen time on computer ($OR=1.04$, $CI\ 95\%=0.84 - 1.27$, $p=.727$) nor either TV ($OR=1.13$, $CI\ 95\% 0.96 - 1.45$, $p=.125$).

Table 3: Multivariate logistic regression unadjusted analysis and adjusted analysis, Odds ratio (OR) with a 95% confidence interval (CI) for in the association between Screen time and Mental health problems for the total sample. All screen time variables included simultaneously in each analysis.

Variable	Unadjusted OR (CI 95 %)	p	Adjusted OR (CI 95 %)	p
Smartphone	1.61 (1.30-1.99)	<.001	1.72 (1.39 – 2.13)	<.001
Computer	1.07 (0.87-1.31)	.51	1.04 (0.84 – 1.27)	.727
TV	1.29 (1.05-1.58)	.014	1.13 (0.96 – 1.45)	.125
Sedentary status			1.25 (1.11 – 1.41)	<.001
Perceived socio-economic status			0.64 (0.58 – 0.70)	<.001

Notes: Dependent variable=Dichotomized with cut off >13 →Mental health problems. Independent variables= Screen time → Smartphone, Computer and TV. OR = Odds ratio. CI= Confidence interval. p = significant level. Model 1= Unadjusted OR without confounders. Model 2= Adjusted OR= controlling for sedentary status and perceived socioeconomic status as confounders.

5.2 Associations between screen time and mental health problems in girls and boys

Furthermore, the sample was stratified into girls and boys (Table 4). The unadjusted results among girls found the smartphone to be associated with mental health problems ($OR=1.32$, $CI\ 95\ \% = 1.00-1.7$, $p=.047$). Thus, high screen time on smartphone increases the odds by 32% for having mental health problems compared to girls with low screen time. In addition, both the computer ($OR=1.47$, $CI\ 95\% 1.14-1.88$ $p=.002$) and the TV ($OR=1.44$, $CI\ 95\ \% =1.13-1.84$,

p.003) was associated with mental health problems among girls. High screen time on computer increases the odds of 47 % for diagnosable mental health problems among girls. In addition, high screen time from watching TV increases the odds by 44% for mental health problems compared to girls with low screen time. Furthermore, the unadjusted analyses reveal no significant results among boys for none of the three screen types.

Finally, the results were adjusted for potential confounders (*Table 4*). It was found among girls with high screen time on smartphone (*OR*=1.42, *CI* 95%=1.06 – 1.89, *p*.019) increase the odds of 42 % of having mental health problems compared to girls with low screen-time. Girls with high screen time on computer (*OR*=1.44, *CI* 95%=1.11 – 1.85, *p*.005) increase the odds by 44 % for mental health problems. In addition, girls with high screen time from watching TV increase the odds by 33 % for having mental health problems (*OR*=1.33, *CI* 95%=1.03– 1.72, *p*.027) compared to girls with low screen time from the TV. However, no significant association was found among boys when adjusting for confounders.

Table 4: Multivariate logistic regression unadjusted analysis and adjusted analysis for the association between Screen time and Mental health problems separately for girls and boys. All screen time variables included simultaneously in each analysis.

Variable	Unadjusted OR (CI 95 %)	p	Adjusted OR (CI 95 %)	p
<u>Girls</u>				
Smartphone	1.32 (1.00-1.75)	.047	1.42 (1.06 – 1.89)	.019
Computer	1.47 (1.14-1.88)	.002	1.44 (1.11 – 1.85)	.005
TV	1.44 (1.13-1.84)	.003	1.33 (1.03 – 1.72)	.027
Sedentary status			1.19 (1.01 – 1.39)	.031
Perceived socio-economic status			0.62 (0.55 - 0.70)	<.001
<u>Boys</u>				
Smartphone	1.30 (0.91-1.84)	.15	1.39 (0.96 – 2.00)	.075
Computer	0.87 (0.60-1.27)	.48	0.82 (0.56 – 1.20)	.312
TV	1.17 (0.80-1.70)	.41	1.05 (0.72 – 1.54)	.800
Sedentary status			1.34 (1.10 – 1.65)	.004
Perceived socio-economic status			0.65 (0.56 – 0.76)	<.001

Notes: Dependent variable= Dichotomized with cut off >13=Mental health problems. Independent variables= Screen time → Smartphone, Computer and TV. OR = Odds ratio. CI= Confidence interval. *p*= significant level. Model 1= Unadjusted OR without confounders. Model 2= Adjusted OR= controlling for sedentary status and perceived socioeconomic status as confounders.

6 DISCUSSION

The study aimed to examine the association between screen time (i.e., smartphone, computer, and TV) and mental health problems among Swedish adolescents and investigate if the association was different for girls and boys. The results indicate that screen time contributes to mental health problems. Furthermore, smartphone use increased the probability of having diagnosable mental health problems among adolescents. However, no association was found for screen time on TV and computer in the leisure time. Therefore, the smartphone was found to be the strongest predictor of mental health problems. Finally, when the sample was stratified, all three screen types (smartphone, computer, and TV) among girls increase the probability of having mental health problems but not among boys.

6.1 Method discussion

The present study used a cross-sectional study design. The study design contributes to the fact that the dependent and independent variables were measured at the same time. According to Setia (2016), the study design allows for investigating associations between the variables of interest and is suitable for population-based surveys (Setia, 2016). However, a limitation of the study design is that it is not possible to conclude anything about the causal relationships, which can be considered as a limitation of the present study. Further studies should consider more robust study designs to assess the direction of the relationship and conclude causality.

6.1.1 *Sample and data collection*

The present study has a large sample size of adolescents from the Swedish population. The response rate of SALVe 2020 was 71 % and is considered acceptable. Söderqvist and Larm (2021) claim that the whole population study with relatively high participation rates is a strength. Therefore, the sample and collected data are well representative among adolescents in county Västmanland. Thus, the results may be generalized to all adolescents in Sweden because of the large sample size (Söderqvist & Larm, 2021). The advantage of using secondary data is the large sample size that otherwise not have been possible to obtain. Hence, the representative sample increases the generalization of the present study results to other adolescents living in Sweden.

The external validity in the present study is considered as good. The present study is considered to have good representativeness since all adolescents in Västmanland in ninth grade and the second year from upper secondary school was invited to participate. It is also noteworthy to consider the non-response of 29 %. It may have caused that adolescent with the worst mental health status did not participate in the survey. Consequently, it may be a risk that youth experiencing mental health problems were not in school, thus not answering the questionnaire. Furthermore, the risk for selection bias can be increased because the individuals that did not participate could have important characteristics for the sample.

Despite this, the sample is still considered to have a sufficient sample size since 71 % completed the survey.

6.1.2 Measurements and statistical analysis

A limitation is the use of a self-reported questionnaire since it allows the participants to under- or over-estimate their answers. There is also a risk that the participants do not answer the questions entirely truthfully (Creswell & Creswell, 2018). Hence, there could be a risk for response bias if the participants give incorrect information in the survey. It could be because some of the questions may be perceived as sensitive, for example, *"I felt so depressed that nothing could cheer me up"*. Therefore, it can be a risk for participants not answering the question or giving a false answer to not reveal the truth. It is also possible that they perhaps have overestimated their screen time and underestimated their mental health. Since the participants may not remember all perceived emotional states from the latest month. It may also have been challenging to estimate the daily screen time.

The SALVe 2020 questionnaire was used as measurement instruments and the survey has been conducted and distributed several times before. In addition, the questionnaire has been used in the previous scientific literature (Hellström et al., 2017; Söderqvist & Larm, 2021). Consequently, it is considered to increase the questionnaire validity since several researchers have used the instrument.

To answer the present study research questions, the Kessler K6 scale was selected to measure mental health problems. It is a globally used instrument with high validity to estimate the population at risk for a diagnosable mental health problem such as anxiety and depression. The scale has been used in international research, which enables good validity of the measurement. Therefore, the instrument is valid and reliable to measure mental illness (Kessler et al., 2002; Kawakami et al., 2020; Kang et al., 2015). The psychological distress scale is originally in English, but the scale has been translated into over 30 languages. The different translated versions have shown good usefulness (Furukawa et al., 2008). The scale may have been affected in the translation into Swedish. However, Furukawa et al. (2008) claim that the scale has been translated into several languages but still been valid. Given this, the translation into Swedish should not have affected the validity of the scale.

The K6 scale is a general screening tool that can capture early indications of potential symptoms of overall mental illness. However, Gilbert et al. (2001) argue that it is important to notice that screening tools do not capture definitively diagnoses nor either a definitive indication. Therefore, other diagnostic tools are needed to determine a clinical diagnose and to conclude mental illness. Thus, screening tools can generate suspicion of mental illness, but a diagnostic tool is required to confirm a mental disease (Gilbert et al., 2001). The fact that the K6 scale is a screening tool means that the tool can identify potential cases of anxiety- and depressive symptoms. However, it cannot be diagnosed as a mental illness even if the results indicate it. Thus, there is a risk that the instrument may identify cases with mental illness even though these individuals may have only reacted to stressful life situations. Kessler et al. (2003) claim that the original K6 scale with the cut off ≥ 13 has a sensitivity of 0.36, specificity of 0.96, and classification accuracy of 0.92. Therefore, the instrument has

low sensitivity (Kessler et al., 2003), which may increase the risk for false identified cases of mental illness. However, the specificity is high and decrease the risk for capture cases that do not have a mental symptom.

Furthermore, the K6 scale has been a useful screening tool in research and can accurately determine potential mental illness. Especially in population-based health surveys to estimate probabilities for serious mental illness, the instrument has shown good accuracy (Furukawa et al., 2003; Kessler et al., 2003; Veldhuizen et al., 2007). In another study, the psychological distress scale has Cronbach's α 0.88 (Cornelius et al., 2013). Kawakami et al. (2020) study had Cronbach's α value of 0.86. In the present study, the Cronbach's alpha value of the index from the scale was 0.91. Field (2018) claims that Cronbach's alpha values over 0.7 are good (Field, 2018). In previous research, the psychological distress scale has been established as reliable when Cronbach's alpha is between 0.89 to 0.92, and the original scale has Cronbach's α value of 0.89 (Kessler et al., 2002). In addition, Ursachi et al. (2015) claim that Cronbach's alpha values of 0.8 to 0.95 indicate very good reliability and internal consistency. Hence, Cronbach's alpha value in the present study indicates high reliability. Therefore, it is considered a strength of the study since the value of the measurement is acceptable. Veldhuizen et al. (2007) also claim the instrument is accurate to identify mental health issues. Consequently, the present study has used a valid measurement for assessing the outcome. It is possible to conclude that the instrument measures what it intends to assess, and it is well calibrated. It contributes to increasing the present study's credibility.

Three questions were selected to measure screen time. Since the primary interest was directed to examine the different types of screens impact on mental health, each screen time variable was dichotomized to make them comparable. The cut-off > 4 hours per day was considered reliable since other studies have applied it (Shiue, 2015; Duncan et al., 2012). Furthermore, previous researcher has similarly measured screen time. For instance, Madhav et al. (2017) measured screen time with the question, on average, how many hours per day do you use a computer outside the school and the same for watching TV.

The multicollinearity was checked between all the included predictors. Hence, the risk for bias in the results considered to be limited since no strong correlations occurred. The logistic regression was considered to generate the most credible results. However, it can also be a limitation to dichotomize all the variables because it may increase the risk of missing important nuances and values. Therefore, the dichotomization may have impacted the association when not considering the whole psychological distress range and screen time ranges. The variables were dichotomized in line with how previous research has done to avoid misleading results. Previous studies have used four hours of screen time as the cut-off for screen time variables (Shiue, 2015; Duncan et al., 2012). In addition, several studies have used the psychological distress scale as dichotomized, with the cut-off > 13 to predict the probability for serious mental health problems (Mitchell & Beals, 2011; Prochaska et al., 2012; Furukawa et al., 2008).

The included confounders were as expected related to the outcome. However, the sedentary status was used to control the real effects of screen time. Since previous studies has found sedentary status to be associated with both screen time and mental health problems

(Tremblay et al., 2011). However, there may be a risk for residual confounding in the results. This could be due to other potential confounders remaining that have not been assessed. According to Walsh et al. (2020) is mental health a multidimensional concept. Thus, it exists several risk factors for adolescent's mental health problems. For example, low social support, drug- and alcohol abuse have not been assessed in the present study. Therefore, it may be a risk due to not measured risk factors impact on the results. However, the primary interest of this study was to examine eventually impact of sedentary behaviour and socioeconomic status affected the association.

Finally, the identified differences among girls and boys in the current study need to be investigated further with other analyses. In the present study, the sample was stratified, which only indicates the different odds ratio separately for girls and boys. It can be considered a limitation since gender was only used as a potential moderator for the association. Hence, it is not possible to conclude eventually statically significant gender differences.

6.1.3 Ethical considerations

The author in the present study was not involved in the data collection of the survey. The use of secondary data contributes to difficulties to control all the ethical considerations. Since the data collection was conducted during school time, there may be a risk that the adolescent felt compelled to participate. The Declaration of Helsinki was followed, and ethical approval was given in the study (Söderqvist & Larm, 2021). Thus, important research ethics is considered to have been met.

In the current study, ethical considerations of importance were to maintain the participants' confidentiality. Since the results are reported on a group level, and no information that makes it possible to identify an individual was collected, the participants were kept anonymous. The current study only used variables in the dataset of relevance to answering the research questions. In addition, the data was kept so that no unauthorized persons had access to the password-protected computer. The data was only used for the present study, and the dataset was deleted from the author's computer afterwards the study was completed. With that in mind, the present study is considered to have met the ethical considerations.

6.2 Results discussion

Nowadays, adolescents are exposed to screen time from multiple devices, such as TV, smartphones, and computers. This study found an association between screen time and mental health problems, which was in line with previous research (Twenge & Campbell, 2018). However, previous research has claimed a clear difference between the different screen types, and they contribute to mental health to varying degrees. Therefore, it has been suggested to examine each activity separately (Twenge & Farley, 2021). Screen time has often been measured without distinguishing the type of screen, including several types of screen activities. In this study, on the other hand, each dimension of screen time was measured

separately to examine the impact on mental health and determine each dimension contribution to mental health.

The present study measured three different screen types. It was expected that screen time contribute negatively to mental health. However, this study results only confirm high screen time on the smartphone to be associated with an increased probability for mental health problems. Interestingly, screen time on TV and computer during leisure time was not related to mental health problems in the total sample, contrary to previous findings from earlier studies. For example, Maras et al. (2015) found that computers and TV have a negative impact on mental health. However, according to Rosen et al. (2014), screen time from the TV is less harmful than social media (Rosen et al., 2014). The reason why the computer does not affect mental health maybe because of the social interactions online. Granic et al. (2014) claim that video gaming can benefit mental health since it can offer social meetings with friends.

The smartphone had a rapid adoption in society, leading to a dramatic increase in its use among adolescents (Twenge et al., 2018). The present study confirms smartphone use of four hours per day to increase the probability of mental health problems. In previous studies, Thomée (2018) states that screen time on smartphones is most harmful to adolescent's mental health. In particular, extensive smartphone use has been associated with depression (Thomée, 2018). The result from the present study also indicates the smartphones as the strongest predictor of mental health problems. Accordingly, a potential explanation for the negative impact on mental health might be because of smartphone use takes time from other activities.

However, the present study cannot confirm TV and computer to have harmful effects on adolescents' mental health. According to Domingues-Montanari (2017) is the TV the most common screen type among younger children. It may explain why the present study did not find the TV to impact mental health. An explanation to why the TV not affected mental health in this sample may be that the TV is stationary, since Twenge et al. (2019) claim that the smartphone is portable and always available (Twenge et al., 2019). The fact that smartphones are portable may increase the screen time. Consequently, screen time on TV may be less harmful because it is not available to the same extent as the smartphone.

Girls report mental health problems to a broader extent (Lager et al., 2012). The present study's results found a high amount of screen time to be a possible explanation. When the sample was stratified, the results show that high screen time on all three screen types (smartphone, computer, and TV) was associated with an increased probability for mental health problems among girls. It is in line with previous findings from Oberle et al. (2020), who found screen time of two hours or more each day was more associated with anxiety- and depressive symptoms among girls than boys.

Moreover, Twenge and Farley (2021) confirm the association stronger among girls than boys (Twenge & Farley, 2021). The present study results indicated that watching TV increased the probability of mental health problems among girls. Similar results have already been found- For instance, Liu et al. (2016) found TV watching for at least two hours each day was related to mental symptoms, such as depression. Another study by Stiglic and Viner (2019) has

established excessive TV-watching as a risk factor for mental health problems, which also was emphasized by Twenge et al. (2018). Thus, the result from this study contributes to similar findings, although these only apply to girls. However, further analyses are required to conclude gender differences. Nevertheless, the results indicate that girls seemed more vulnerable to screen time which may be a potential explanation to why girls suffer from mental health problems to a broader extent than boys.

Twenge et al. (2018) claim that adolescents with high screen time on electronic communication (social media, texting, and gaming) replace time with non-screen activities such as physical activity, homework, and personal social interactions. In addition, high screen time was associated with decreased mental health. Adolescents with lower screen time and spending more time on non-screen activities were found to have better mental health status (Twenge et al., 2018). The present study confirms the smartphone to have negative impact on adolescent's mental health. In addition, the computer and TV indicate to be harmful among girls. These results may be explained by the displacement hypothesis. Kraut et al. (1998) claim that extensive screen time increases the risk to displace time for interpersonal relationships (Kraut et al., 1998). The theoretical framework hypothesizes that all screen time has a negative impact on mental health because screen time displaces time for health-promoting activities (Neuman, 1988). Thomée (2018) states that nowadays, adolescents interact digitally. Rosen et al. (2014) argue for increased risk for mental problems when personal interactions are replaced by digital interactions (Rosen et al., 2014). Hence, important cognitive functions for mental health are replaced. In conclusion, the explanation why screen time is harmful to adolescent's mental health could be when activities such as physical activity are replaced in favour of screen activities. Further research is recommended to conclude that it is the displacement of non-screen activities like in-person social interaction causing the increase in mental health problems.

There has been a shift in the usage of the screens. In the past, televisions were the most common screen type used for entertainment. While today it is primarily internet and online communication which constitutes of screen time. Concerning the change in the use of screen devices some aspects of screen time can be beneficial to mental health. Twenge and Campbell (2018) argue for more research studying how screen time contributes to psychological well-being (Twenge & Campbell, 2018). Therefore, a suggestion for further research is to examine the social benefits from screen time related to social media and online communication to investigate eventually positive aspects to mental health.

The contribution from screen time may also be beneficial for mental health. In the scientific literature, it exists both positive and negative impact from screen time on mental health. Therefore, it may also exist positive aspects from screen time contributing to psychological benefits. According to Valkenburg and Peter (2007), can communication on internet with friends contribute to mental well-being. The stimulation hypothesis explains the health benefits of screen time. The theory suggests that online communication positively impacts mental health since the quality of current friendships can be strengthened by messaging on the internet. Hence, there is evidence for communication online with existing friendships can increase adolescence mental well-being. However, the displacement hypothesis argues that internet usage displaces time for "offline" relationships (Valkenburg & Peter, 2007). Thomée

(2018) also claim that health-related behaviours risk to being displaced with screen time. In addition, the frequent usage of social media on smartphones increases the risk for adverse mental health development (Thomeé, 2018).

The current study can form the basis for further interventions targeting mental health. Since the results found extensive screen time on the smartphone to be harmful to mental health. The current recommendations of daily screen time among adolescents are two hours each day during leisure time (Tremblay et al., 2016). Saunders and Vallance (2017) claim that it is necessary to reduce screen time among adolescents to prevent negative consequences. This study found screen time of four hours per day on the smartphone to be a strong predictor for mental health problems among adolescents. Accordingly, this study suggests that policymakers need to consider establishing guidelines for reasonable screen time in an attempt to reduce screen time among adolescents.

6.3 Relevance to public health and future research

The present study contributes to broadening the knowledge about the association between screen time and mental health problems. This study adds important information about how different types of screens affect mental health. The results found especially the smartphone to be a risk factor for adolescent's mental health problems. In the field of public health, it is also of interest to discover if there are some positive aspects from screen time. Suggestions for future research are to evaluate positive impact of screen time to clarify eventually benefits on mental health.

The results indicate that high screen time on all three screen types (smartphone, computer, and TV) was associated with an increased probability of mental health problems among girls. These findings need to be investigated further to conclude gender differences. In addition, more research is necessary to draw valid conclusions, using others analyses and study designs to claim causality. The present study results can form the basis for further research and interventions. Even more, importantly may be to develop guidelines for adolescent screen time usage since no Swedish guidelines were found. General guidelines for the amount of screen time can be a preventative intervention to reduce harmful effects on youth mental health. The present study hopefully adds valuable information to the existing knowledge in the field.

7 CONCLUSIONS

This study adds valuable knowledge about the smartphone as risk factor for mental health problems. The results provide broaden evidence for how different screen devices contribute to mental health, as only the smartphone increased the probability of mental health problems in this sample. Since the amount of screen time is modifiable, it is relevant to highlight the need of developing Swedish guidelines to reduce harmful effects from screen time among adolescents. The increasing trend of mental health problems is a widespread public health issue, and the results can be of interest for policymakers, school staff, and caregivers. In order to implement guidelines limiting screen time may contribute to preventing the development of mental health problems.

The findings also indicate that girls are negatively affected by screen time to a greater extent than boys. Among girls all three screen types (smartphone, computer, and TV) was associated with increased probability for mental health problems. In contrast, no significant results were found among boys. Therefore, high amount of screen time from multiple screens can be an explanation to why girls suffer from mental health problems to a wider extent than boys which could be considered in further research.

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APPENDIX; QUESTIONNAIRE, SALVE 2020

Ungefär hur ofta under den senaste månaden har du...

Sätt ett kryss per rad.

	Hela tiden	För det mesta	En del av tiden	En liten del av tiden	Inte alls
... känt dig nervös?	☐	☐	☐	☐	☐
... känt hopplöshet?	☐	☐	☐	☐	☐
... känt dig rastlös eller orolig?	☐	☐	☐	☐	☐
... känt dig så deprimerad att inget har kunnat muntra upp dig?	☐	☐	☐	☐	☐
... känt att allt varit en ansträngning?	☐	☐	☐	☐	☐
... känt dig värdelös?	☐	☐	☐	☐	☐

Hur många timmar per dag använder du i genomsnitt dator/läsplatta på fritiden (inte i skolan)?

Markera med en ring på tidslinjen för att visa antal timmar.

_____→

Timm ar 0 1 2 3 4 5 6 7 8 9 10 Mer än 10

Hur många timmar per dag ser du i genomsnitt på TV/TV-program (även streaming tjänster)?

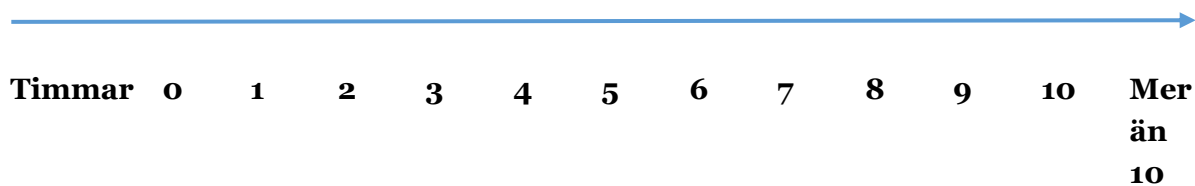
Markera med en ring på tidslinjen för att visa antal timmar.

_____→

Timmar 0 1 2 3 4 5 6 7 8 9 10 Mer än 10

Hur många timmar per dag använder du i genomsnitt din mobiltelefon?

Markera med en ring på tidslinjen för att visa antal timmar.



Föreställ dig samhället som en stege. Om du tänker på din familjs ekonomi i jämförelse med samhället i stort, var skulle du placera din familj på stegen?

(7) Högst upp, mest pengar

(6)

(5)

(4)

(3)

(2)

(1) Längst ner, minst pengar

Hur mycket sitter du under ett normalt dygn?

(4) 10 timmar eller mer

(3) 7–9 timmar

(2) 4–6 timmar

(1) Mindre än 4 timmar



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